MTP F26 · TEAM C · RADIO TEAM → ELECTRONICS TEAM

Suggested hardware for the transceiver

From Radio Team (Santiago, Andrian, Marcos) **To** Carlos, Michele, Bikash, Guillem

Version 0.7 · Wed 21 Oct 2026

New in v0.7 (21 Oct): the carrier is routed. Details in The carrier board.

- **Pi 3 A+ HAT carrier, rev A:** every net is routed. DRC shows **0 violations, 0 unconnected, 0 schematic-parity issues**.
- **Fab files are ready** in `hw/teamc-carrier/fab/`: gerbers, drill, placement file for assembly, assembly PDF and BoM. The zip is ready to upload to a JLCPCB-class fab.
- **Real 3D renders** with every part body, including the HAT sitting on the Pi.
- **For you:** a short review list (5 points in `hw/DESIGN_NOTES.md`), then the order in early November.

v0.6 (19 Oct): the carrier existed in KiCad as a draft.

- **SP4 breakout: ready for the lab.** Single-sided, 31 × 37.5 mm, DRC clean, gerbers and 1:1 copper PDFs in `hw/sp4-breakout/fab/`. It's needed for the first bench tests (T0/T2).
- **Pi 3 A+ HAT carrier: draft for you.** The schematic is complete and independently reviewed. The board was placed with the power path and SPI routed (the rest is now done, v0.7).
- **Parts chosen:** TPS61022 boost, Si2333DDS FETs, the Pi 3B's ideal diode, MCP73831 charger, MAX17048 fuel gauge, TLV75533 radio LDO, SP0504 ESD arrays and a 5 A fuse. The controls are panel-mounted on JST-XH leads.
- **Corrections:** the SP4 is **14.5 × 18 mm**, with 8 castellations on one edge plus 2 GND half-holes. The antenna is the **Ebyte TX2400-FPC-2509**, with the Molex as fallback.
- **Fab:** the carrier is drawn for a **JLCPCB-class fab** (0.2 mm rules); the breakout stays lab-friendly.
- **Cost: ≈ €102 per box**, from the full BoM.

v0.5 (18 Oct): the micro became a Raspberry Pi 3 (the team's request; the 3 A+ is suggested). The radio side did not change. On the board:

- The carrier becomes a **HAT-style board** on the Pi's 40-pin header: SP4, radio LDO, 5 V boost, fuel gauge, charger, switches and LEDs.
- A **5.1 V boost rated for at least 2.5 A** now feeds the Pi all the time. It replaces the small boost and switch that only powered the stick.
- The stick uses the Pi's **own USB port**, so PIO-USB, the stick switch and the USBLC6 are gone.
- The Pi has no ADC, so a **MAX17048 fuel gauge** reads the cell instead of ADC3 and the ideal diode.
- The Pi's **WiFi/BT chip** is disabled in software or desoldered. That is our working assumption; Atenea Q22 asks.
- Cost goes to about **70–90 € per box**. The full analysis is on the Team C Pi 3 Switch page.

v0.5.1 (18 Oct, later): the block diagram shows the HAT boundary, the microSD and the GPIO numbers; the pin map adds the mode switch's second GPIO (16), the fuel-gauge alert (25) and the charger status (26).

Earlier: v0.4.4 added the TX/RX role switch and the secured U.FL plug. v0.4.3 added low-battery countermeasures. v0.4.2 moved SRI/MRM to 20 dBm hopping on the same hardware. v0.4 chose one Ebyte E01-ML01SP4 and one flex omni per box.

The radio baseline is 2.4 GHz with the nRF24L01+. Under our split, the electronics team picks the micro and designs the board. This page is the radio team's **suggestion**, built around every hardware rule in the competition PDF, to save you time. It is not a decision. Please challenge it.

In short

1. **Micro module: Raspberry Pi 3, Model A+ suggested** (Model B works if we already own them). It runs Linux on a microSD card and has native USB host ports. Its WiFi/BT chip is **disabled in software or desoldered**, which is our working assumption until Atenea Q22 is answered.
2. **Radio: one Ebyte E01-ML01SP4 per box**, unchanged: a genuine nRF24L01+ with PA and LNA, SMD **14.5 × 18 mm**, 8 castellations at 1.27 mm plus 2 GND half-holes, and an IPEX antenna socket. **VCC is 2.0–3.6 V only**, so it keeps its own 3.3 V regulator from the cell, never the Pi's 3V3 pin. The same module goes from the bench (on the breakout) to the final boxes.
3. **Board: a 2-layer HAT carrier** on the Pi's 40-pin header, **routed in KiCad, rev A, DRC clean, gerbers ready** (below). It carries the SP4, the radio LDO, the 5.1 V boost that feeds the Pi, an ideal diode, the fuel gauge, the charger, ESD protection, and JST-XH leads to the panel controls. The antenna is the Ebyte TX2400-FPC-2509 flex, stuck on a box wall and plugged into the module, so there is no RF circuitry on our board.
4. **Power: one protected 18650 Li-ion cell** (3000 mAh or more) on the box floor, charged on board through a charge-only USB-C port. The Pi and radio draw about 2 W, so one cell lasts about 4–5 h with the A+. For longer, wire a second holder in parallel into the same battery lead.
5. **Cost: ≈ €102 per device** from the BoM (`hw/BOM.csv`), against the 150 € cap. That includes the Pi, the microSD, panel controls, mechanics and the lab PCB at €15.

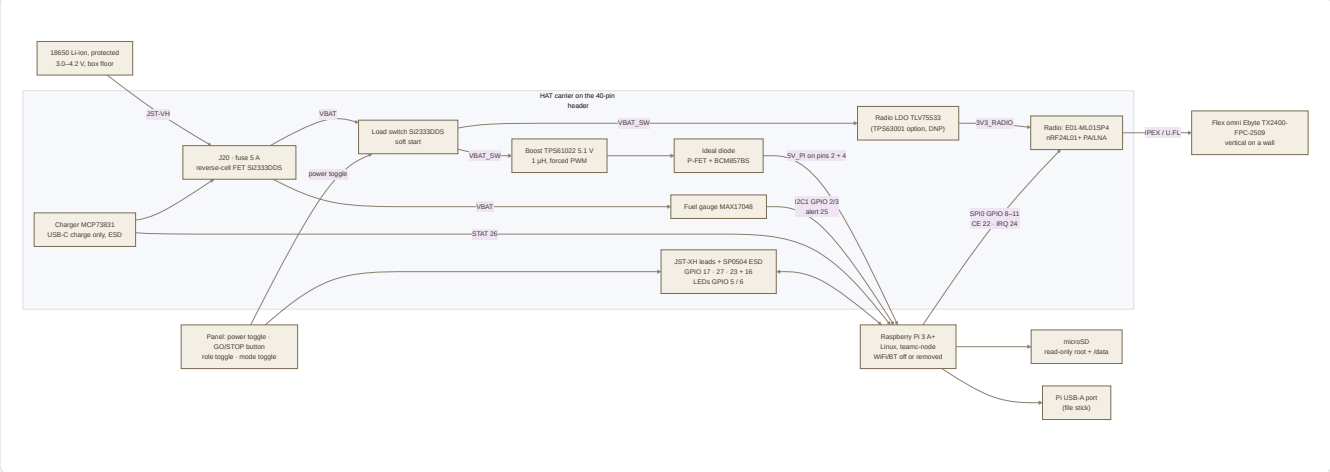
The rules this design answers

RULE (COMPETITION PDF)	WHAT IT FORCES ON THE BOARD
One and only one module including a microprocessor (the rules name "Raspberry..." as an example); no single commercial module embedding the entire system	The Pi 3 is the only micro. The radio, charger, boost and fuel gauge contain no CPU.
No chip implementing WiFi, Bluetooth, ZigBee, 2G–5G	The Pi 3 carries one (CYW43455 on the A+/B+, BCM43438 on the B). **Working assumption:** disabled for good (`dtoverlay=disable-wifi`, `disable-bt`, `rfkill`) or desoldered; **Atenea Q22** asks which one the professors require. Still banned: Pico W / Pico 2 W, ESP32 boards, Arduino Nano 33 BLE/IoT, Uno R4 WiFi, Nano ESP32.
Read/write the file from/to a USB memory device: 5 min setup while still wrapped in foil, 3 min to upload afterwards	A Pi USB-A port at a box wall (or a short internal extension to a panel socket). The microSD card holds the file, so storage is no longer a constraint. The stick can be removed during the round.
Fits entirely in the 15 × 15 × 7 cm IKEA 365+; only connectors, switches and buttons may stick out	Pi 3 A+ 65 × 56 mm, about 30 mm tall with the HAT; the 18650 lies beside it on the box floor (its holder, ≈ 77 mm, is longer than the HAT). The antenna sits inside: a 25 × 9 mm flex omni (Molex fallback 35 mm), vertical, as high as possible on a wall.

RULE (COMPETITION PDF)	WHAT IT FORCES ON THE BOARD
Operates autonomously	A battery and a power switch. Nothing plugged in during a round.
At least one Tx/Rx switch or button; a pilot light that blinks in Tx/Rx	Panel pushbutton for GO/STOP, a **TX/RX role switch** (the professor picks the sender after loading; Atenea Q8), and an activity LED. The box is transparent, so the LED can stay inside.
Extra switches allowed (e.g. network mode), but **no switch indicating the position in the network**	A 3-position mode switch SRI / MRM / NM (two GPIOs). Nothing that encodes a node number.
Drop test from table height to concrete (before MRM)	Pi on M2.5 standoffs; HAT held by four M2.5 × 11 mm standoffs; **microSD taped or glued** (the Pi 3 slot holds it by friction only); battery strapped; antenna glued down; **U.FL plug secured**. The panel button and toggles are nut-clamped in the box wall, so impacts land on the wall rather than on solder joints, and their JST-XH housings are glued.
Designed to self-recover from interruptions	Linux boots in 10–30 s instead of the Pico's 0.3 s. Target ≤ 10 s, hardware watchdog through systemd, read-only root so a power cut cannot corrupt the card. Sim T19: one 30 s reboot in MRM still delivers the file in 90.8 s.
ESD and RF exposure compliance	SP0504 ESD arrays at every panel lead and at the USB-C (CC1, CC2, VBUS); TVS on the 5 V and battery rails. RF power is small (≤ 10 mW e.i.r.p.; ≤ 100 mW e.i.r.p. while hopping in SRI/MRM).
Modules shown at Quick Mode are final	The QM prototype uses the same Pi 3 model and the same E01-ML01SP4 as the final board, and the same flex antenna. Q22 must be answered before 6 Nov.
Unit cost ≤ €150; lab-made PCB counts €15; everything (screws, foam, wires, owned boards) goes in the BoM	A Pi we already own still counts at market price.

Block diagram

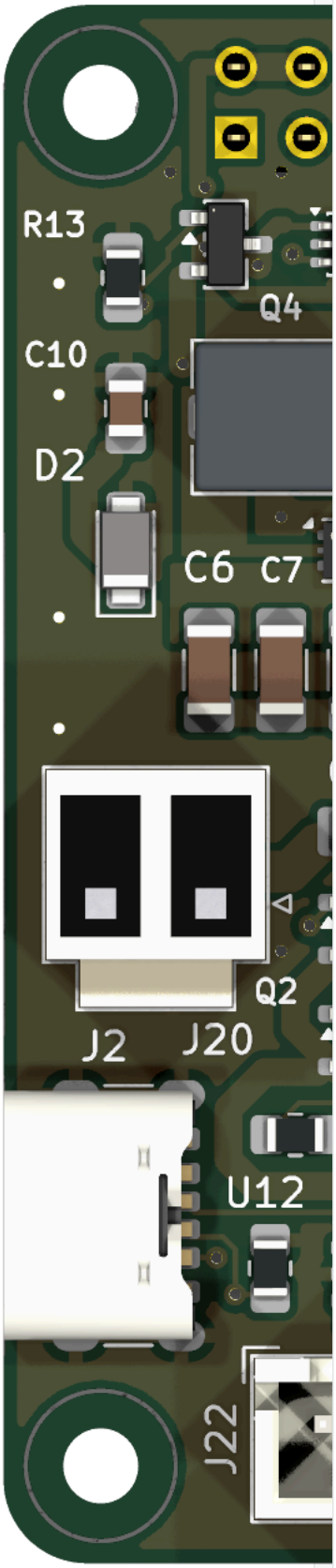
The only fast data path is SPI to the radio, at 31 kB/s (250 kbps). USB file transfer happens only before and after a round.



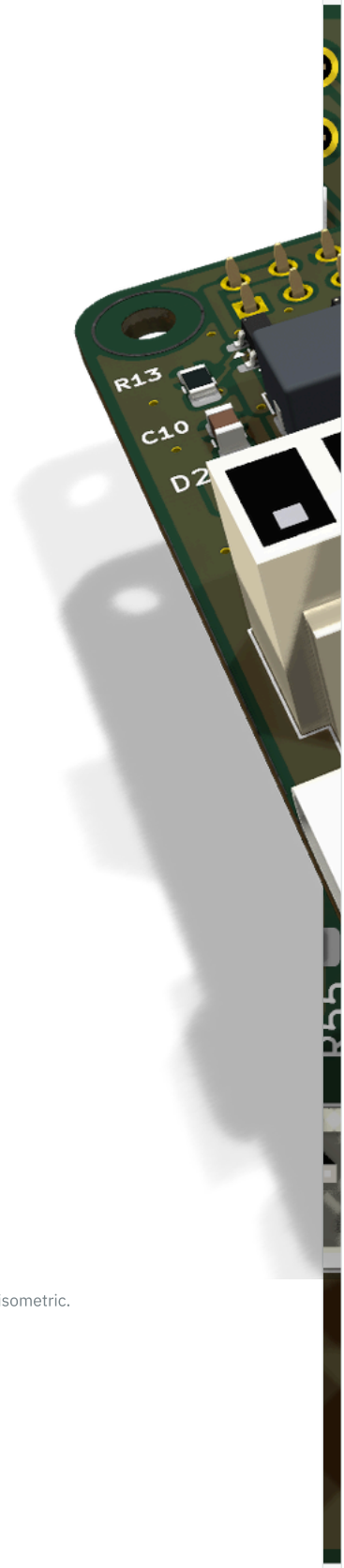
The battery, the antenna and the Pi's own parts (microSD, USB port) sit outside the HAT. The software that runs on this hardware, `teamc-node`, is written and tested against an emulated radio; it has not run on a Pi yet (Team C Protocol Proposal v0.4.1).

The carrier board (KiCad rev A, 21 Oct)

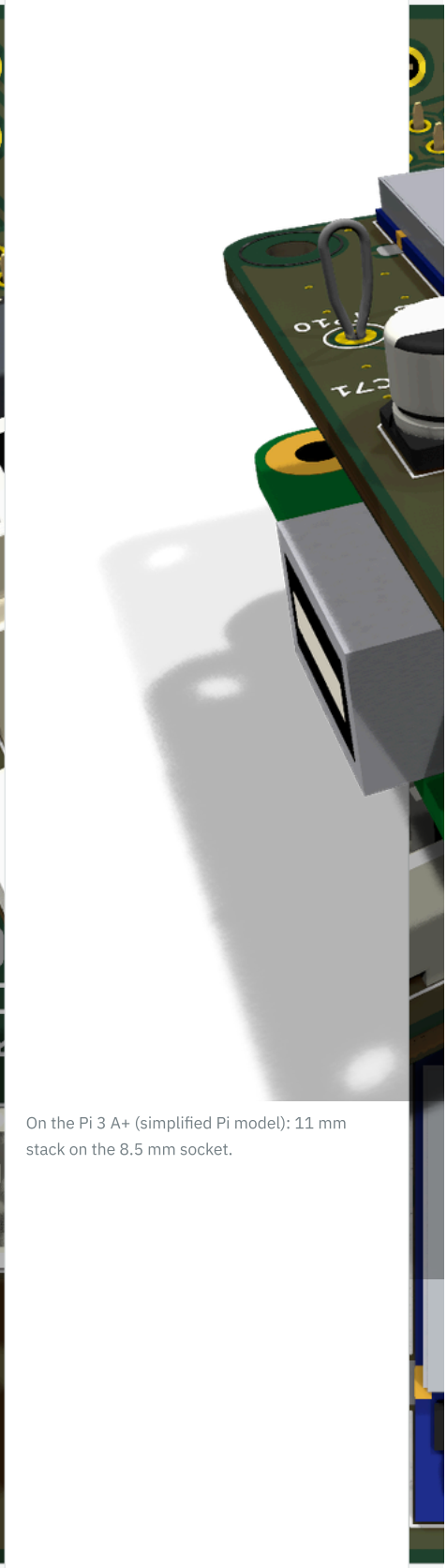
Designed and routed by the radio team for you to review and order. Everything lives in the repo under hw/. Read `hw/DESIGN_NOTES.md` first; its section "State of the carrier board on 21-Oct" lists what to check. The schematic files are generated from `hw/teamc-carrier/sheets/*.py`, so edit those and run `python3 hw/scripts/build_carrier.py`.



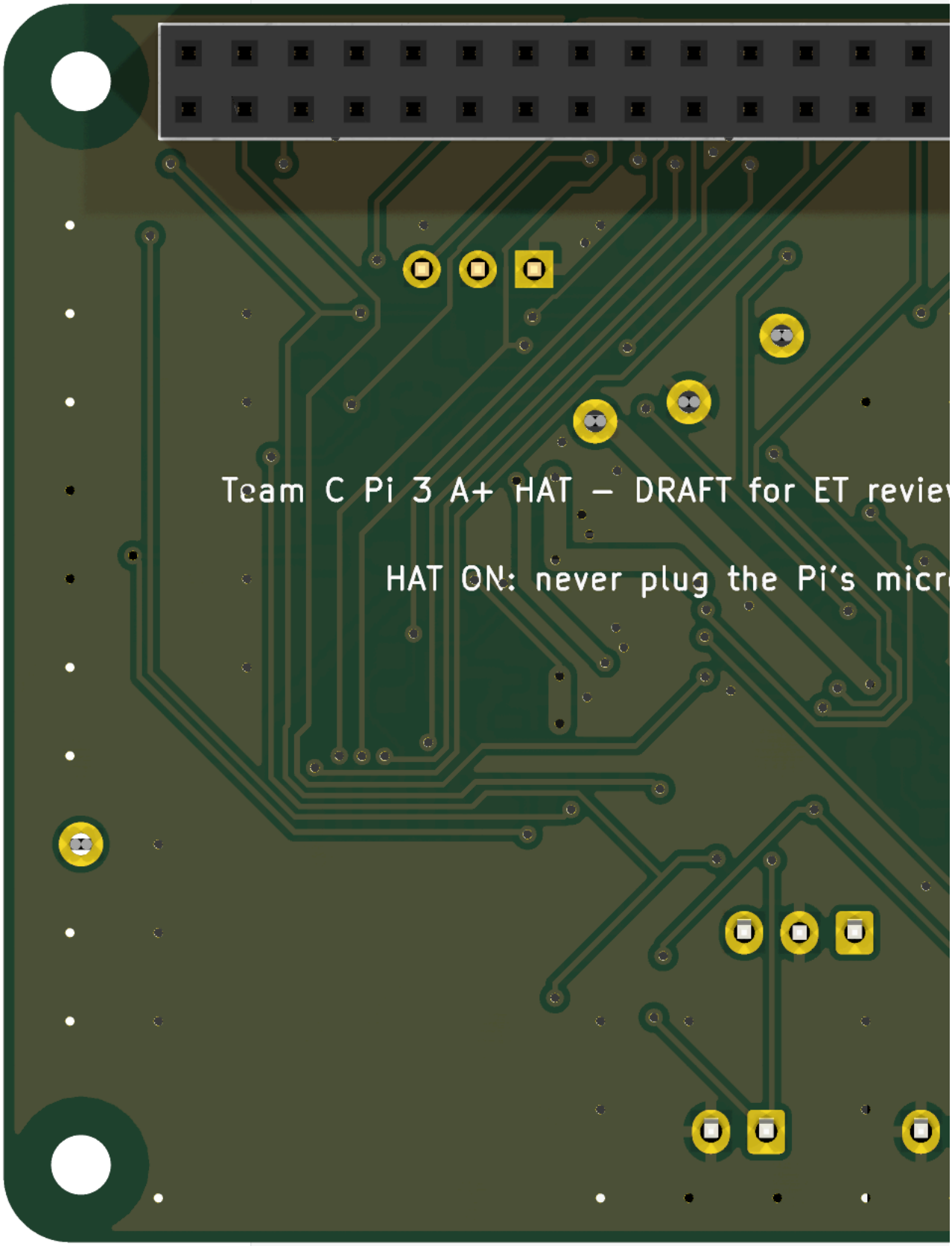
Carrier, top: 65 × 56 mm, the A+ outline and holes. DNP parts (the buck-boost option) show as bare pads.



Carrier, isometric.



On the Pi 3 A+ (simplified Pi model): 11 mm stack on the 8.5 mm socket.



Carrier, bottom: the 2 × 20 socket and the ground pour.

ITEM	STATE	WHERE / NOTES
SP4 breakout	ready for the lab	Single-sided, 31 × 37.5 mm. 1 × 8 header in the usual nRF24 order plus a 1 × 2 bench-supply header; 10 µF + 100 nF + 100 µF. ERC/DRC 0. Gerbers, drill and 1:1 copper PDFs (plain and mirrored) in `hw/sp4-breakout/fab/`. Measure one module (14.5 mm) before etching.
Carrier schematic	complete	4 sheets (power, pi_header, radio, ui), ERC 0 errors / 0 warnings, pin map checked against `ui.c`, the radio README and pinout.xyz. An independent review found no blockers; its fixes are in, with the table in DESIGN_NOTES. PDF: `hw/teamc-carrier/teamc-carrier_schematic.pdf`.
Placement	done	Boost laid out as TI's example, next to header pins 2/4. Battery path J20 → fuse → Q1 → Q2 on the left edge, charger and USB-C below it. Panel leads on the bottom edge. SP4 bottom-right, ≥ 45 mm from the Pi's wireless can, no tracks under it on either layer.
Routing	done	**DRC: 0 violations, 0 unconnected, 0 schematic-parity issues**, at all severities. The power path is routed by script with the boost copper areas and wide tracks. The rest was routed by Freerouting, and a short finishing pass handled the pins it could not reach. Ground pours are on both layers, with 71 GND vias. 696 tracks, 146 vias.
What the finishing pass changed	please review	The GND pins of the MAX17048 and the TPS63001 are tied to their exposed pads. In the DNP buck-boost option, L2 is turned 90° (TI's layout) and C15 180°, and one 3V3_RADIO via moved. Some ground pockets reach the main pour through short GND tracks. All of it is in DESIGN_NOTES, with 5 review points.
Fab rules	fab, not lab	0.2 mm clearance, 0.25 mm signals, 0.15 mm at the two 0.5 mm-pitch chips, plated vias. That fits a JLCPCB-class 2-layer fab (≈ €1/board, 1–2 weeks). At the lab-safe 0.3 mm this board does not route.
Gerbers for the carrier	ready	`hw/teamc-carrier/teamc-carrier_gerbers.zip` (gerbers and drill) uploads as-is: 2 layers, 1.6 mm, HASL, 1 oz. In `fab/`: the placement file in JLCPCB's columns (DNP left out), the assembly PDF, and copper PDFs per layer.
Silkscreen and 3D	done	59 reference designators fit on the silkscreen at 1.0 or 0.8 mm. The other 26 have no room and are on the assembly drawing only. 3D models for every part, including the SP4 and the MAX17048 (our own).

> **Rule for everyone:** with the HAT switched on, **never plug the Pi's micro-USB**. The A+ has no reverse blocking on that input, so our boost would push current into the supply. It is on the board's silkscreen too.

Micro module options

The team wants a Pi 3. The table ranks the Pi 3 variants and keeps the alternatives for comparison.

MODULE	WIRELESS CHIP	DRAW AT 5 V	PRICE	VERDICT	NOTES
Raspberry Pi 3 Model A+	CYW43455, off or removed	~0.30–0.40 A	~€27	suggested	Lowest draw and smallest Pi 3 (65 × 56 mm); one USB port wired straight to the SoC; no Ethernet or hub chip to power. Built until at least Jan 2030.
Raspberry Pi 3 Model B	BCM43438, off or removed	~0.35–0.45 A	~€37	if owned	85 × 56 mm, 4 USB ports behind a LAN9514 hub. Fine if the team already has them.
Raspberry Pi 3 Model B+	CYW43455, off or removed	~0.45–0.60 A	~€40	highest draw	Shortest battery life of the three.
Compute Module 3+ (Lite)	None	~0.3 A	~€27–33	fallback	Same processor with no wireless chip at all, if Atenea says the chip must go and desoldering fails. Needs a SODIMM carrier we would design. Stock is patchy.
Raspberry Pi Zero (non-W)	None	~0.1–0.2 A	~€5–10	alternative	Linux too, no wireless question, a third of the draw. Single core, so real-time timing is harder; the USB host needs an OTG adapter.
Raspberry Pi Pico 2 (RP2350)	None	~0.03–0.05 A	~€5	previous plan	Cheapest, 0.3 s reboot, exact timing. Its port (PIO-USB, flash rules) stays in git history.
Pico W / Pico 2 W, ESP32, Nano 33 BLE/IoT, Uno R4 WiFi, Pi Zero W, Pi 4/5	—	—	–	banned	On-board WiFi/BT chip, and nothing to gain over the Pi 3.
Micro + radio combos (Feather RP2040 RFM69, Moteino)	—	—	–	avoid	Risks the "single commercial module embedding the entire system" rule.

Prices are estimates seen online in October 2026. Get real quotes before ordering.

Parts per device

The parts on the KiCad draft, grouped by function. Line-by-line MPNs, suppliers and prices (excluding VAT, 19 Oct) are in `hw/BOM.csv`. Electronics owns the final choice and the sourcing. Check stock before ordering: the TPS61022 and the Si2333DDS showed 0 at DigiKey; the AO3401A fits the same footprint.

FUNCTION	PARTS	WHY	€ / DEVICE
Micro module	Raspberry Pi 3 Model A+ (ordered)	Owned boards count at market price	27.00
Storage	SanDisk Ultra 32 GB A1 microSD (ordered)	Linux + files; read-only root	6.00
Radio + antenna	E01-ML01SP4 + TX2400-FPC-2509 (ordered); 100 µF + 10 µF + 100 nF at pad 1; 33 Ω SPI series; pull-ups	Same module from the breakout to the box	4.31
Battery	Protected 18650 ≥ 3000 mAh, MPD BH-18650-PC holder (THT, strapped)	On the box floor; JST-VH lead	10.94
Battery path	JST-VH J20, 5 A fast fuse, Si2333DDS reverse-cell FET, Si2333DDS load switch (soft start), JST-XH for the power toggle, TVS	Header pins 2/4 bypass the Pi's polyfuse; ≤ 40 mΩ at −2.5 V near an empty cell	2.21
5 V for the Pi	TPS61022 + Würth 1 µH (11.5 A), 3 × 22 µF + 22 µF + 2 × 100 µF; ideal diode (P-FET + BCM857BS, the Pi 3B circuit); TVS	The A+ has no ideal diode; 5.10 V	7.51
Charger + USB-C	MCP73831-2 (452 mA), GCT USB4125 charge-only, CC 5.1 kΩ, SP0504 ESD (CC1/CC2/VBUS), LED, 2N7002 STAT level shift	Charges with the switch off	1.88
Radio supply	TLV75533 LDO (+ TPS63001 buck-boost footprint, DNP)	Never the cell, never the Pi's 3V3	0.44
Battery sense	MAX17048 fuel gauge (I2C 0x36, alert to GPIO 25)	The Pi has no ADC	4.05
Panel controls	E-Switch PS1024ABLK button, 100SP1T1B1M1QEH × 2 (power, role), 100SP3T1B1M1QEH (mode, ON-OFF-ON)	Nut-clamped in the box wall	8.70
UI on the board	JST-XH leads, SP0504 ESD, 10 k / 1 k / 100 nF per input, 2 LEDs	Inputs active low; LEDs active high	1.87
Header + debug	Harwin 2 × 20 socket (11 mm stack), UART header, test points		3.75
Harness + mechanics	JST housings and crimps, wire, M2.5 standoffs and screws, IKEA 365+ box, glue, tape, solder	Everything goes in the BoM (rules)	8.56
PCB	Carrier HAT, 2-layer	Counted at the lab's fixed €15 (rule); a JLCPCB-class fab is ≈ €1/board	15.00
Total per device (all items at market price)			**≈ 102**

FUNCTION	SUGGESTED PART	WHY	≈ € / DEVICE
Micro module	Raspberry Pi 3 Model A+ (B if owned)	See above. Owned boards count at market price	27–37
Storage	microSD 16 GB, high-endurance or industrial	Holds Linux and the files. Read-only root, small data partition	5–8
Radio module	1 × Ebyte E01-ML01SP4 (nRF24L01P + PA/LNA, SMD, IPEX). **Not** E01C-* (clone chip)	Chip-controlled PA/LNA, so only SPI + CSN + CE + IRQ. Same module from the bench (on a breakout) to the final box	≈ 4–6
Antenna	Molex 146153-0100 flex omni (34.9 × 9 mm, 100 mm cable, U.FL)	Plugs straight into the module. Vertical polarisation, mounted high on a wall	≈ 3
5 V for the Pi	Boost 5.1 V, ≥ 2.5 A out from 3.0–4.2 V (TPS61088-class + inductor; a ready-made 5 V 3 A module for the bench)	The Pi needs a steady 5 V; it flags undervoltage below 4.63 V. Near the 3.0 V cutoff the boost pulls about 2 A from the cell	3–5

FUNCTION	SUGGESTED PART	WHY	≈ € / DEVICE
Battery sense	MAX17048 fuel gauge on I2C1	The Pi has no ADC. Reads cell voltage and state of charge	2
Protection	TVS + reverse-polarity protection on the 5 V feed; TVS on the button lines	Feeding the header pins bypasses the Pi's own input protection	0.6
Battery	1 × protected 18650 Li-ion (3000 mAh or more) with solder tabs, or holder + strap; a second cell in parallel if needed	Steel can is robust in the drop test; rechargeable (sustainability report)	6–8
Charger	MCP73831 at 500 mA + USB-C receptacle, charge only	Charges a 3000 mAh cell overnight (6–7 h). Nothing is plugged in during operation	1.2
Controls	Slide power switch, panel GO/STOP pushbutton, TX/RX role switch, 3-position mode switch (on-off-on), 2 LEDs	Required by the rules	2.5
Radio supply	3.3 V LDO (AP2112K-3.3 or TLV75533, dropout ≤ 0.1 V at 150 mA) from the cell + 100 µF / 10 µF / 100 nF at the module	SP4 VCC is 2.0–3.6 V (abs. max 3.6 V); PA/LNA modules are sensitive to supply noise; handles the ~120 mA TX bursts. Never the Pi's 3V3 pin	1
PCB	2-layer HAT-style carrier, 65 × 56 mm (A+ outline), 40-pin header	ETSETB lab = €15 in the BoM (rule); an external fab is ~€1 per board but adds 1–2 weeks	1–15
Mechanics	M2.5 standoffs (58 × 49 mm pattern), base plate, cable ties, hot glue, tape for the SD card, IKEA 365+ box	Drop test; the BoM must list them too	5–7
Total per device (verify with quotes)			**≈ 70–90**

Power budget

Current draw

Pi 3 A+ at 5 V	≈ 0.30–0.40 A, one core busy-waiting for the radio ESTIMATE
SP4 TX	≈ 120 mA peak DATASHEET
SP4 RX	≈ 26 mA DATASHEET
USB stick (Pi port)	50–100 mA, only during load/save ESTIMATE
From the cell	≈ 0.5–0.8 A; up to ≈ 2 A at boot near 3.0 V

Battery sizing

Usable energy of one 3000 mAh cell is about 3.0 Ah × 3.6 V × 85 % boost efficiency ≈ 9.2 Wh. With the radio's ~0.15 W that gives:

- Pi 3 A+, 1.7–2.2 W: **4.2–5.4 h**
- Pi 3 B, 2.0–2.4 W: 3.8–4.6 h
- Pi 3 B+, 2.4–3.2 W: 2.9–3.8 h

A competition day of ~4 h powered on fits the A+ on one cell, barely. Use two cells in parallel, or switch the box off between rounds (boot ≤ 10 s). Measure the real draw at T0.

Low battery

The radio LDO from the cell is still fine: a clean 3.3 V down to a cell of about 3.4 V, about 2.95 V at the 3.0 V cutoff, inside the SP4's 2.0–3.6 V. The new weak point is the **5 V boost**: if it sags below 4.63 V the Pi throttles, and lower it crashes.

- **Battery sense:** MAX17048 on I2C1 (no ideal diode needed any more). The firmware also reads the Pi's undervoltage flag (`vcgencmd get_throttled`) and logs it.
- **Firmware:** thresholds (3.5 / 3.4 / 3.2 V) re-measured at T0 under the Pi's load, since the cell sags more at 0.5–1 A. Below the lowest, fsync received data after every burst.
- **T0** measures the radio output and the Pi's current at 3.6, 3.3 and 3.0 V.
- **Boost:** TPS61022 with a 1 µH / 11.5 A inductor (peak ≈ 5.7 A at 3.0 V in, 2.5 A out); 100 µF + 100 nF at the Pi's 5 V pins. The cell path (fuse + two Si2333DDS) drops ≈ 0.15 V at 2 A.
- **No hardware under-voltage cut-off:** the cell's protection board and the software thresholds do it, so switch the box off after a round.
- **Radio fallback:** the board has a TPS63001 buck-boost footprint (DNP). Fit it instead of the LDO's 0 Ω link if T0 shows more than 2 dB less output at 3.0 V.

For the sustainability report

The Pi 3 box uses about 7× the energy of the Pico design (≈ 2 W against ≈ 0.3 W). In its favour: the Pi is reflashable, repairable and reusable after the course, and the HAT unplugs. We can measure watts per round directly with the fuel gauge.

Build stages

STAGE	BOARD	GOAL	TARGET
1 · Bench	Pi 3 + E01-ML01SP4 on the **SP4 breakout** (`hw/sp4-breakout/fab/`, ready) wired to the header, radio 3.3 V from a bench supply, Pi on its own 5 V supply	Radio tests T0 (output power, Pi noise in our band) and T2 (frame gap on Linux); USB-stick read/write on the Pi's port	as soon as parts arrive
2 · Quick Mode	Perfboard HAT, **same Pi model and module part numbers**, external power allowed	One error-free line from a file in < 2 min, first attempt	week 6–7
3 · PCB rev A	The KiCad carrier rev A (routed, gerbers ready): review, order from a JLCPCB-class fab (1–2 weeks), assemble, box, battery	Autonomous, drop-tested, range tests T4/T5	order by early Nov · week 8–9
4 · PCB rev B	Fixes from rev A	Pretest (4 Dec) fully packaged	week 10–11

Risks and how we cover them

RISK	MITIGATION
The professors require the WiFi/BT chip to be removed	Atenea Q22 before Quick Mode. Try desoldering on one spare board now. Last resort: Compute Module 3+ (no chip).
SD card corrupted by a power cut or the drop test	Read-only root (overlayfs) with a small data partition, high-endurance card, card taped or glued in its slot.
Linux reboot after an "attack" or a brown-out takes 10–30 s	Boot ≤ 10 s target, hardware watchdog via systemd, resume from the last acknowledged line (sim T19: 90.8 s in MRM with a 30 s reboot).
The 5 V boost sags at low charge and the Pi crashes	Boost sized for 2 A input at 3.0 V, bulk capacitance, fuel gauge warning, a second cell in parallel if T0 shows trouble.
Pi clocks (6 × 400 MHz = 2400 MHz, HDMI) add noise in our band	HDMI off, SP4 at the board edge away from the SoC, antenna high on the wall; T0 logs RPD hits per hop channel with the Pi idle and busy, and drops a bad channel.
The foil-wrapped setup makes the USB port hard to reach	A Pi USB port facing a side wall, or a short internal extension to a panel socket. Atenea Q6 asks whether the foil may be opened at the port.
Clone radio chips (Si24R1)	Buy only the plain E01-ML01SP4 part number from Ebyte's store or a distributor; T0/T2 check the chip.
SMD module needs a breakout before any PCB exists	The breakout is drawn and ready for the lab (single-sided, lab-safe rules); the HAT takes over in stage 3.
The carrier's 0.5 mm-pitch chips (TPS61022, MAX17048) and 0.2 mm rules are beyond a hand-made lab board	A JLCPCB-class fab for the carrier; stencil + hot air (or the fab's assembly) for the two chips. The fuel gauge has a plan B: a MAX17048 breakout module on headers.
Someone plugs the Pi's micro-USB while the HAT is on	Operating rule + silkscreen; the ideal diode protects our boost, but the A+ has no blocking on its own input.
A protected cell trips on the inrush at power-on	Soft start on the load switch (≈ 1 ms); bench-check with the real cell.
Drop test pulls the HAT off the header or ejects the SD card or battery	HAT soldered or on four standoffs; SD taped; strapped cell; drop rev A on purpose in week 9.

What we need from electronics

1. **An owner for the carrier:** review rev A in the KiCad GUI. The 5 points are in `hw/DESIGN_NOTES.md`: the ground links, the thermal copper at the charger, tidying Freerouting's longer tracks, the box dimensions and panel leads, and one SP4 measured. Then place the fab order. To have the boards in week 8–9, order in early November.
2. **Make the SP4 breakout in the lab** from `hw/sp4-breakout/fab/` for the first bench test, and measure one module first (14.5 mm).
3. **Review the carrier schematic and the part choices** (the PDF, and the review table in DESIGN_NOTES). Two to confirm: MODE tied to VIN (forced PWM), and the cell on the box floor with a second cell paralleled in the harness.
4. **Check stock** for TPS61022, Si2333DDS (AO3401A fallback), the 5 A 1206 fuse and MAX17048.
5. Try desoldering the WiFi/BT chip on one spare A+ (Atenea Q22), then check that the Pi still boots and that SPI and USB work.
6. The IKEA box's inside dimensions, so the panel holes and the antenna position can be fixed.

Radio details (interface pins, test plan, 20 dBm hopping in SRI/MRM): Team C Radio Brief v0.11. Firmware and USB: Team C Protocol Proposal v0.4.1. Pi 3 analysis: Team C Pi 3 Switch. Rules quoted from *MTP F26 – Program Competition Rules – Network Upgrade III + Sustainability*.

KiCad files, BoM and design notes: github.com/SantiBarrios2002/mtp-f26-teamc/tree/main/hw (carrier rev A – for ET review).

Sources checked 18 Oct 2026: Pi 3 A+ · Pi 3 B · Pi power and USB documentation · measured Pi 3 current · Ebyte E01-ML01SP4 manual · nRF24L01+ Product Specification.